HDOC-32

Aim:

To write a menu-driven C program using separate functions to:

1. Input two variable-length 2D arrays and verify that **matrix addition is commutative**, i.e. A + B = B + A.
2. Input two variable-length 2D arrays and demonstrate that **matrix multiplication is generally not commutative**, i.e. A × B ≠ B × A.
3. Exit the program.

Task1: Algorithm:

1. Start the program.
2. Display the menu.
3. Read the user's choice.
4. If choice is **1**:
    - Read rows and columns.
    - Create two VLAs A and B of the same size.
    - Input both matrices.
    - Calculate A + B.
    - Calculate B + A.
    - Compare the results.
    - Display that matrix addition is commutative if they are equal.
5. If choice is **2**:

- To calculate **both A×B and B×A and compare them directly**, take two square matrices of the same order.
- Calculate A×B.
- Calculate B×A.
- Compare and display the results.
- If they differ, this demonstrates that multiplication is not commutative.

6. If choice is **3**, terminate the program.
7. For any other choice, display an error message.
8. Repeat until Exit is selected.

Task 2: Test case table:

Test Case	Choice	No. of Rows	No. of Columns	Matrix A	Matrix B	Expected Output	Status
1	1	2	2	1 2; 3 4	5 6; 7 8	A+B = B+A = [6 8; 10 12] → Addition is commutative	Pass
2	1	2	3	1 2 3; 4 5 6	6 5 4; 3 2 1	A+B = B+A = [7 7 7; 7 7 7] → Addition is commutative	Pass
3	2	2	2	1 2; 3 4	5 6; 7 8	AB = [19 22; 43 50], BA = [23 34; 31 46] → AB ≠ BA	Pass
4	2	3	3	1 0 0; 0 2 0; 0 0 3	1 2 0; 0 1 0; 0 0 1	AB ≠ BA → Multiplication is not commutative	Pass
5	3	—	—	—	—	Displays "Exiting program..."	Pass

Task3 :C PROGRAM:

```
#include <stdio.h>
/* Function to input a matrix */
void inputMatrix(int rows, int cols, int matrix[rows][cols])
{
   int i, j;
   for (i = 0; i < rows; i++)
   {
      for (j = 0; j < cols; j++)
      {
         scanf("%d", &matrix[i][j]);
      }
   }
}
/* Function to display a matrix */
void displayMatrix(int rows, int cols, int matrix[rows][cols])
{
   int i, j;
   for (i = 0; i < rows; i++)
   {
      for (j = 0; j < cols; j++)
      {
         printf("%d\t", matrix[i][j]);
      }
      printf("\n");
   }
}
/* Function for matrix addition */
void addMatrix(int rows, int cols,
               int A[rows][cols],
               int B[rows][cols],
               int result[rows][cols])
{
   int i, j;
   for (i = 0; i < rows; i++)
   {
      for (j = 0; j < cols; j++)
      {
         result[i][j] = A[i][j] + B[i][j];
      }
   }
}
/* Function to compare two matrices */
int compareMatrix(int rows, int cols,
                  int A[rows][cols],
```

```
                int B[rows][cols])
{
   int i, j;
   for (i = 0; i < rows; i++)
   {
      for (j = 0; j < cols; j++)
      {
         if (A[i][j] != B[i][j])
            return 0;
      }
   }
   return 1;
}
/* Function for multiplication of square matrices */
void multiplyMatrix(int n,
                    int A[n][n],
                    int B[n][n],
                    int result[n][n])
{
   int i, j, k;
   for (i = 0; i < n; i++)
   {
      for (j = 0; j < n; j++)
      {
         result[i][j] = 0;
         for (k = 0; k < n; k++)
         {
            result[i][j] += A[i][k] * B[k][j];
         }
      }
   }
}
int main()
{
   int choice;
   do
   {
      printf("\n========== MATRIX OPERATIONS ==========\n");
      printf("1. Prove Matrix Addition is Commutative\n");
      printf("2. Prove Matrix Multiplication is Not Commutative\n");
      printf("3. Exit\n");
      printf("Enter your choice: ");
      scanf("%d", &choice);
      switch (choice)
      {
         case 1:
```

```
{
    int rows, cols;
    printf("\nEnter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);
    int A[rows][cols];
    int B[rows][cols];
    int AB[rows][cols];
    int BA[rows][cols];
    printf("\nEnter elements of Matrix A:\n");
    inputMatrix(rows, cols, A);
    printf("\nEnter elements of Matrix B:\n");
    inputMatrix(rows, cols, B);
    /* Calculate A + B */
    addMatrix(rows, cols, A, B, AB);
    /* Calculate B + A */
    addMatrix(rows, cols, B, A, BA);
    printf("\nA + B:\n");
    displayMatrix(rows, cols, AB);
    printf("\nB + A:\n");
    displayMatrix(rows, cols, BA);
    if (compareMatrix(rows, cols, AB, BA))
    {
        printf("\nA + B = B + A\n");
        printf("Hence, Matrix Addition is COMMUTATIVE.\n");
    }
    else
    {
        printf("\nA + B != B + A\n");
    }
    break;
}
case 2:
{
    int n;
    printf("\nEnter the order of the square matrices: ");
    scanf("%d", &n);
    int A[n][n];
    int B[n][n];
    int AB[n][n];
    int BA[n][n];
    printf("\nEnter elements of Matrix A:\n");
    inputMatrix(n, n, A);
    printf("\nEnter elements of Matrix B:\n");
    inputMatrix(n, n, B);
    /* Calculate A x B */
    multiplyMatrix(n, A, B, AB);
```

```
            /* Calculate B x A */
            multiplyMatrix(n, B, A, BA);
            printf("\nA x B:\n");
            displayMatrix(n, n, AB);
            printf("\nB x A:\n");
            displayMatrix(n, n, BA);
            if (!compareMatrix(n, n, AB, BA))
            {
               printf("\nA x B != B x A\n");
               printf("Hence, Matrix Multiplication is NOT COMMUTATIVE.\n");
            }
            else
            {
               printf("\nFor these particular matrices, A x B = B x A.\n");
               printf("However, matrix multiplication is not "
                     "commutative in general.\n");
            }
            break;
         }
         case 3:
            printf("\nExiting program...\n");
            break;
         default:
            printf("\nInvalid choice! Please enter 1, 2 or 3.\n");
      }
   } while (choice != 3);
   return 0;
}
```

Task4: OUTPUT:

```
═══════ MATRIX OPERATIONS ═══════
1. Prove Matrix Addition is Commutative
2. Prove Matrix Multiplication is Not Commutative
3. Exit
Enter your choice: 1

Enter number of rows and columns: 2
2

Enter elements of Matrix A:
1
2

3
4

Enter elements of Matrix B:
5
6
7
8

A + B:
6       8
10      12

B + A:
6       8
10      12

A + B = B + A
Hence, Matrix Addition is COMMUTATIVE.
```

```
========== MATRIX OPERATIONS ==========
1. Prove Matrix Addition is Commutative
2. Prove Matrix Multiplication is Not Commutative
3. Exit
Enter your choice: 1

Enter number of rows and columns: 2
3

Enter elements of Matrix A:
1
2
3
4
5
6

Enter elements of Matrix B:
6
5
4
3
2
1

A + B:
7       7       7
7       7       7

B + A:
7       7       7
7       7       7

A + B = B + A
Hence, Matrix Addition is COMMUTATIVE.
```

```
========== MATRIX OPERATIONS ==========
1. Prove Matrix Addition is Commutative
2. Prove Matrix Multiplication is Not Commutative
3. Exit
Enter your choice: 2

Enter the order of the square matrices: 2

Enter elements of Matrix A:
1
2
3
4

Enter elements of Matrix B:
5
6
7
8

A x B:
19      22
43      50

B x A:
23      34
31      46

A x B ≠ B x A
Hence, Matrix Multiplication is NOT COMMUTATIVE.
```

```
========== MATRIX OPERATIONS ==========
1. Prove Matrix Addition is Commutative
2. Prove Matrix Multiplication is Not Commutative
3. Exit
Enter your choice: 2

Enter the order of the square matrices: 3

Enter elements of Matrix A:
1
0
0
0
2
0
0
0
3

Enter elements of Matrix B:
1
2
0
0
1
0
0
0
1

A x B:
1       2       0
0       2       0
0       0       3

B x A:
1       4       0
0       2       0
0       0       3

A x B ≠ B x A
Hence, Matrix Multiplication is NOT COMMUTATIVE.

========== MATRIX OPERATIONS ==========
1. Prove Matrix Addition is Commutative
2. Prove Matrix Multiplication is Not Commutative
3. Exit
Enter your choice: 3

Exiting program...

┌──(kali㉿kali)-[~/hdoc32]
└─$
```